Two Types of Bijections and Cantor's Logical Fallacy

Qi Jin *

October 18, 2026

Abstract

There are two types of bijections between infinite sets: partial field bijection and full field bijection. The bijection used to compare the cardinalities of sets are partial field in nature, exemplified by the bijection between even numbers and natural numbers. Cantor failed to distinguish between these two different types of bijections, which led to logical fallacies and erroneous conclusions, such as $|N| < |P(N)|$ and $|R| = |P(N)|$. This paper logically demonstrates Cantor's logical errors.

Keywords: infinite set, partial field bijection, full field bijection, Cantor, logical fallacy
MSC2020: 03A05

1 Introduction

Let $f : A \leftrightarrow B$ denote a full field bijection between sets A and B. If A and B are infinite sets, partial field bijection also exist.

For two infinite sets A and B, let $C \subset A$ and $D \subset B$ be their dynamic proper subsets. A dynamic proper subset is defined as a proper subset whose number of elements can increase indefinitely.

The mapping $f : C \leftrightarrow D$ constitutes a partial field bijection between A and B, which is denoted by $f_p : A \leftrightarrow B$. Thus,

$$f_p : A \leftrightarrow B = f : C \leftrightarrow D.$$

*Independent Researcher, Guangzhou, China. ORCID: 0009-0002-9700-4587.
Email: 871685523@qq.com. No funding was received for this work.

2 Bijection between N and EN

The bijection between the natural numbers (denoted by N) and the even numbers (denoted by EN) is a partial field bijection, rather than a full field bijection.

When N = 2, the set of even numbers contains only the element 2, whereas the set of natural numbers contains two elements, 1 and 2. Therefore, a full field bijection cannot be established.

If a full field bijection cannot be established when N = 2K, then it cannot be established when N = 2(K+1) either. This is because the set of even numbers increases by only one element (2K+2), while the set of natural numbers increases by two elements (2K+1 and 2K+2). Consequently, establishing a full field bijection becomes even more impossible.

Can a complete mapping ever be achieved by continuing this process infinitely? Obviously not. With each step of mapping, new unmapped natural numbers are generated, and their quantity keeps increasing, as shown in Figure 1. Continuing this process infinitely makes it impossible to ever complete the mapping. This is exactly like a person trying to catch their own shadow. A person can certainly reach the position where the shadow was a second ago, but can they ever catch the shadow by continuing to chase it forever? Obviously not, because the shadow is moving forward at the exact same time.

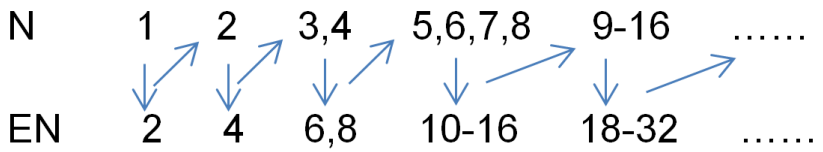


Figure 1 Bijection between N and EN

Achilles can catch up with the turtle, but a person can never catch their own shadow. While the distance between Achilles and the turtle is shrinking, the numerical gap here is expanding. Originally, if even numbers are mapped to even numbers, odd numbers are left without any corresponding elements. The gap of the odd numbers is merely shifted to the end, relying on the concept of infinity to conceal this discrepancy. However, the gap is never actually eliminated; this is logically completely flawed. It commits the fallacy of equivocation by substituting an ongoing, unending process with something that can be completed.

Doing = Finished?
Doing for ever = Can Finish?

If one does not accept that the mapping between natural numbers and even numbers is a partial field bijection, it will lead to the following self-contradiction.

3 Bijection between N and P(N)

Cantor proved that a bijection cannot be established between the set of natural numbers $\mathbb{N}$ and its power set $P(\mathbb{N})$[1]. The logic of his proof by contradiction is as follows:

Assume that there exists a bijection $f : \mathbb{N} \leftrightarrow P(\mathbb{N})$.

Construct the set $D = \{n \in \mathbb{N} \mid n \notin f(n)\}$. That is, D contains all natural numbers that are not mapped to subsets containing themselves.

Since D is a subset of $\mathbb{N}$, $D \in P(\mathbb{N})$. According to the assumption that f is a surjection, there must exist some d in $\mathbb{N}$ such that $f(d) = D$.

Now the question is: Is d in D?

If $d \in D$, then by the definition of D, $d \notin f(d) = D$, which is a contradiction.

If $d \notin D$, then by the definition of D, $d \in f(d) = D$, which is also a contradiction.

Conclusion: The assumption does not hold. There is no bijection between $\mathbb{N}$ and $P(\mathbb{N})$, meaning $|\mathbb{N}| < |P(\mathbb{N})|$.

This proof contains a logical flaw, as it implicitly assumes that the natural numbers terminate at N, rather than increasing infinitely. One could use N+1 to correspond to D. In the bijection between natural numbers and even numbers, numbers greater than N, such as N+2, N+4, ..., 2N, are allowed to appear.

Since natural numbers can be put into a bijection with even numbers, one could simply leave the odd numbers idle and only use the even numbers to establish the correspondence. By establishing bijection:

$$f : EN \leftrightarrow N \leftrightarrow P(N) \Rightarrow f' : EN \leftrightarrow P(N),$$

let $D = f(1)$. Since the other odd numbers are not used, no contradiction will arise.

One can also prove by contradiction that there is no one-to-one correspondence between the natural numbers and the even numbers, as follows:

Assume that a one-to-one correspondence has been established between the set of natural numbers N and the set of even numbers EN.

Then the natural numbers N+1, N+2, ..., 2N would have no corresponding even numbers.

Therefore, the assumption does not hold, and there exists no one-to-one correspondence between the natural numbers and the even numbers.

By using the same method that establishes a one-to-one correspondence between the natural numbers and the even numbers, one can similarly establish a one-to-one correspondence between P(N) and N. The method is as follows:

As shown in Table 1, odd numbers exist between the even numbers—that is, in the gaps between the even numbers in the table. One need only shift the even numbers to the left, leaving the gaps on the right, thereby establishing a one-to-one correspondence.

N	1	2	3	4	5	6	7	8	...
EN	←	2	←	4	←	6	←	8	...
N	1	2	3	4	5	6	7	8	...
EN	2	4	6	8	...	...	...	...	...

Table 1: Bijection between N and EN

Applying the same method to $P(N)$ and N, where $P(N) = 2^N$, as shown in Table 2: shift the natural numbers to the left and leave the gaps on the right, and a one-to-one correspondence is thereby established.

P(N)	1	2	3	4	...	8	...	16	...
N	1	←	←	2	←	3	←	4	...
P(N)	1	2	3	4	...	8	...	16	...
N	1	2	3	4	...	...	...	...	...

Table 2: Bijection between P(N) and N

In summary, consider the following two propositions:

Proposition 1: A bijection can be established between N and P(N).

Proposition 2: A bijection can be established between N and EN.

These two propositions must either hold simultaneously or fail simultaneously. It is impossible for only one to hold while the other does not. One must acknowledge the existence of two types of bijections: partial field bijection (the bijection between natural numbers and even numbers) and full field bijection (the type of bijection that Cantor negated using proof by contradiction).

Proof by contradiction cannot be used to prove that two sets have different cardinalities, because the bijection required to compare cardinalities is a partial field bijection, not a full field bijection. Proof by contradiction only demonstrates that a full field bijection cannot be established; it does not negate the existence of a partial

field bijection. We have

$$f_p : A \leftrightarrow B \neq f : A \leftrightarrow B.$$

Therefore, $|N| < |P(N)|$ has not been truly proven.

4 Bijection between R and P(N)

In addition, Cantor's proof of the bijection between the real numbers and the power set of natural numbers[2] also contains logical errors.

In the third step of the proof, an injection $g : [0, 1) \to P(N)$ is constructed.

For any real number x in the interval, its unique standard binary expansion is selected. The convention is to use the representation that does not end with an infinite sequence of 1s, denoted by $x = 0.C_1C_2C_3....$ The subset g(x) is defined as the set of all natural numbers n such that C_n equals 1.

Proof of injection: If two real numbers x_1 and x_2 are not equal ($x_1 \neq x_2$), their standard binary expansions must differ in at least one digit. The corresponding natural number will belong to only one of the two sets, meaning $g(x_1) \neq g(x_2)$. Therefore, g is an injection.

Real numbers represented in standard binary form correspond to the points of semi-equal division on the number line, as shown in Figure 2. They account for only a very small fraction of the real numbers. Infinite decimal expansions (the numbers between these division points) cannot be represented in standard binary form at all. Irrational numbers, such as $\pi - 3$ and $\sqrt{2} - 1$, cannot be exactly represented as decimals, whether in base-10 or base-2.

Cantor substituted the entire set of real numbers on $[0, 1)$ with this very small fraction of real numbers, thereby obtaining an incorrect result. Therefore, $|R| = |P(N)|$ has also not been proven.

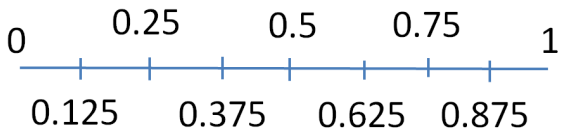


Figure 2 Binary Decimal

References

[1] Georg Cantor. *Ueber eine elementare Frage der Mannigfaltigkeitslehre*. Jahresbericht der Deutschen Mathematiker-Vereinigung, 1891, **1**, pp. 75–78.

[2] Georg Cantor. *Ein Beitrag zur Mannigfaltigkeitslehre. Journal für die reine und angewandte Mathematik*, 1878, **84**, 242–258.